Lecture 3

Stress and energy flux

An oscillator conserves its energy when its Lagrangian has no explicit time dependence. A field can also carry energy from one place to another. Time translation gives a local balance law. The energy in a fixed rectangle is constant on a time interval exactly when the net outward flux through its faces vanishes throughout that interval.

Objectives. *We derive the scalar stress tensor from translations and prove the energy balance on rectangular regions. We compute the energy carried out by a massless pulse and compare the stress tensor with metric variation.*

By the end of this lecture you will be able to:

- Derive the scalar translation current, with its off-shell divergence.
- Compute the energy change in a rectangle from its boundary flux.
- Relate the scalar canonical tensor to the inverse-metric variation.

3.1 Translation currents

We use the coordinates and metric of Lecture 1 on $M = \mathbb{R} \times \mathbb{R}^d$ or $M = \mathbb{R} \times \mathbb{T}^d$. A density is *autonomous* if it has no explicit dependence on x. The momentum current remains $p^\mu = \partial L / \partial(\partial_\mu \phi)$.

Definition 3.1 (Canonical stress tensor)**.** For a scalar field ϕ, the *canonical stress tensor* is the array

$$T^\mu{}_\nu = p^\mu \partial_\nu \phi - \delta^\mu_\nu L[\phi], \qquad \mu, \nu = 0, \dots, d. \tag{3.1}$$

Its first index is the direction of flow. The second index specifies the translation. We raise or lower indices with η.

We find the current for each translation using Theorem 2.4. For a characteristic Q^1 with $\delta_Q L = \partial_\mu B^\mu$, that theorem gives $j^\mu = p^\mu Q^1 - B^\mu$ and $\partial_\mu j^\mu = -E[\phi] Q^1$ before imposing the field equation.

Let e_ν be the coordinate vector in the x^ν direction. For an autonomous density, take the translated field $\phi_\varepsilon(x) = \phi(x + \varepsilon e_\nu)$. Its characteristic is $Q^1 = \partial_\nu \phi$. Since the density has no explicit coordinate dependence, the chain rule gives $\delta_Q L = \partial_\nu L$. Thus we can take $B^\mu = \delta^\mu_\nu L$, and the current is

$$j^\mu = p^\mu \partial_\nu \phi - \delta^\mu_\nu L = T^\mu{}_\nu.$$

This fixed-coordinate field change corresponds to the coordinate translation $x'^\rho = x^\rho - \varepsilon \delta^\rho_\nu$, with $\phi'(x') = \phi(x)$.

We now use the real scalar density with a potential $V \in C^2(\mathbb{R})$:

$$L = \frac{1}{2} \partial_\mu \phi \, \partial^\mu \phi - V(\phi), \qquad E[\phi] = -\big(\Box \phi + V'(\phi)\big). \tag{3.2}$$

Its momentum current is $p^\mu = \partial^\mu \phi$. The canonical tensor with both indices lowered is

$$T_{\mu\nu} = \partial_\mu \phi \, \partial_\nu \phi - \eta_{\mu\nu} L. \tag{3.3}$$

This tensor is symmetric: $T_{\mu\nu} = T_{\nu\mu}$. Symmetry here follows from the scalar density, rather than from the definition of the canonical tensor for an arbitrary field theory.

Definition 3.2 (Energy, momentum and energy flux). For the scalar density (3.2), the *energy density* ρ_{en}, the *momentum density* $\mathcal{P}^i$ and the *energy flux* f^i are

$$\rho_{\text{en}} = T^0{}_0 = \frac{1}{2}\phi_t^2 + \frac{1}{2}|\nabla\phi|^2 + V(\phi), \qquad \mathcal{P}^i = T^{0i}, \qquad f^i = T^{i0} = -\phi_t \partial_i \phi. \tag{3.4}$$

For a bounded spatial region Ω, its *energy* at time t is

$$E_{\text{en},\Omega}(t) = \int_\Omega \rho_{\text{en}}(t,x)\, \mathrm{d}^d x.$$

The same definition applies to all space when the integral is finite.

The flux component f^i measures signed transport in the positive x^i direction. Since the tensor is symmetric, $\mathcal{P}^i = f^i$ for this scalar field. The mixed component has the opposite spatial sign: $T^0{}_i = -\mathcal{P}^i$. In the absence of space derivatives, $\rho_{\text{en}} = \frac{1}{2}\dot\phi^2 + V(\phi)$ is the oscillator energy. For $V(\phi) = \frac{1}{2}m^2\phi^2$, it is the Klein–Gordon energy computed in Lecture 1. If $V \geq 0$, then $\rho_{\text{en}} \geq 0$.

3.2 Local balance and boundary flux

For a rectangle we can state the boundary terms using its faces. Let

$$\Omega = \prod_{i=1}^{d} [a_i, b_i], \qquad a_i < b_i, \qquad \Omega_i = \prod_{j \neq i} [a_j, b_j].$$

The notation $\mathrm{d}^{d-1}x_{\widehat{i}}$ means integration in all coordinates except x^i. For $d = 1$, a face integral means evaluation at the endpoint. The outward normal to the face $x^i = b_i$ is $+e_i$. At $x^i = a_i$ it is $-e_i$. The *net outward flux* is therefore

$$\mathcal{F}_\Omega(t) = \sum_{i=1}^{d} \int_{\Omega_i} \left[f^i(t,x)\big|_{x^i=b_i} - f^i(t,x)\big|_{x^i=a_i} \right] \mathrm{d}^{d-1}x_{\widehat{i}}. \tag{3.5}$$

The sign refers to the outward direction on each face, even when f^i itself is positive on both faces.

Theorem 3.3 (Scalar stress and energy balance). *For $V \in C^2(\mathbb{R})$ and $\phi \in C^2(M, \mathbb{R})$ with density (3.2), translations give (3.1), with*

$$\partial_\mu T^\mu{}_\nu = -E[\phi]\partial_\nu\phi = \big(\Box\phi + V'(\phi)\big)\partial_\nu\phi. \tag{3.6}$$

If ϕ solves $\Box\phi + V'(\phi) = 0$, then

$$\partial_t \rho_{\mathrm{en}} + \partial_i f^i = 0. \tag{3.7}$$

On a fixed rectangle $\Omega \subset \mathbb{R}^d$, its energy satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t} E_{\mathrm{en},\Omega}(t) = -\mathcal{F}_\Omega(t), \qquad E_{\mathrm{en},\Omega}(t_1) - E_{\mathrm{en},\Omega}(t_0) = -\int_{t_0}^{t_1} \mathcal{F}_\Omega(t)\,\mathrm{d}t. \tag{3.8}$$

On $\mathbb{T}^d$, the total energy is constant.

Proof. The translation argument above identifies the current. We verify its divergence directly and then integrate one coordinate at a time.

First, differentiating the scalar density along x^ν gives

$$\partial_\nu L = \partial^\mu\phi\,\partial_\nu\partial_\mu\phi - V'(\phi)\partial_\nu\phi.$$

The product rule in (3.1) and equality of mixed second derivatives now give

$$\begin{aligned}\partial_\mu T^\mu{}_\nu &= (\Box\phi)\partial_\nu\phi + \partial^\mu\phi\,\partial_\mu\partial_\nu\phi - \partial_\nu L\\ &= \big(\Box\phi + V'(\phi)\big)\partial_\nu\phi.\end{aligned}$$

This proves (3.6). On a solution its right-hand side is zero. Taking $\nu = 0$ gives (3.7), since $T^i{}_0 = T^{i0} = f^i$.

Next, ρ_{en} and f^i are C^1. On each compact time interval, the parameter-integral theorem recalled in Lecture 1 permits $\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega \rho_{\mathrm{en}} = \int_\Omega \partial_t\rho_{\mathrm{en}}$. For each spatial index i, Fubini's theorem and the fundamental theorem of calculus give the following identity, with no sum over i:

$$\int_\Omega \partial_i f^i\,\mathrm{d}^d x = \int_{\Omega_i}\int_{a_i}^{b_i} \partial_i f^i\,\mathrm{d}x^i\,\mathrm{d}^{d-1}x_{\widehat{i}} = \int_{\Omega_i}\big[f^i\big|_{b_i} - f^i\big|_{a_i}\big]\,\mathrm{d}^{d-1}x_{\widehat{i}}.$$

Summing and using (3.7) proves the derivative identity in (3.8). The flux is continuous in t. Integration in time proves the finite-time identity.

Finally, on $\mathbb{T}^d$ integrate over $[0, 2\pi]^d$. Opposite face values agree by periodicity, so every difference in (3.5) vanishes. ∎

For a solution on $\mathbb{R}^d$ whose first derivatives ϕ_t and $\nabla\phi$ vanish outside one fixed compact spatial set throughout $[t_0, t_1]$, and with $V(\phi) = 0$ outside that set, a containing rectangle has zero boundary flux. Its energy is the total energy, so the total energy is constant on that interval. For more general decay, the cutoff proof in Lecture 2 applies to the energy current under the density and spatial-flux integrability hypotheses stated there.

3.3 A pulse leaving an interval

We compute the energy remaining in an interval while a massless pulse leaves through its right endpoint. Take $d = 1$, $V = 0$ and

$$\phi(t, x) = F(x - t), \qquad F(s) = \begin{cases}(1 - s^2)^3, & |s| < 1,\\ 0, & |s| \geq 1,\end{cases} \qquad I = [-2, 2], \quad t \geq 0.$$

The function F and its first two derivatives vanish at $s = \pm 1$, so F is C^2 on $\mathbb{R}$. Writing $s = x - t$, the derivatives are $\phi_t = -F'(s)$, $\phi_x = F'(s)$ and $\phi_{tt} = \phi_{xx} = F''(s)$. Thus this field solves the massless equation. Its support at time t is $[t-1, t+1]$: the pulse travels right at speed one.

Figure 3.1 shows the support moving across I. The two sloping lines mark $x = t - 1$ and $x = t + 1$. The vertical lines mark the endpoints of I. The pulse reaches $x = 2$ at $t = 1$ and finishes crossing at $t = 3$.

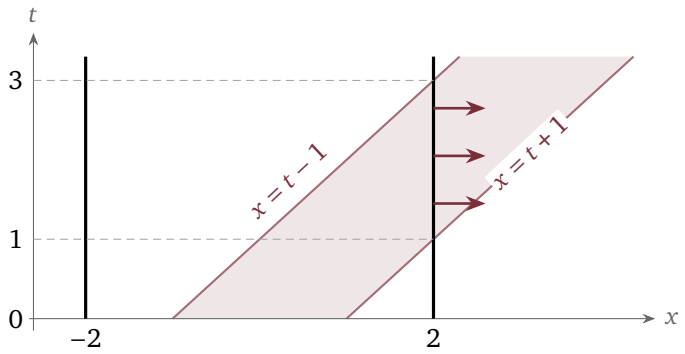


Figure 3.1: The shaded support of the right-moving pulse intersects the fixed interval $[-2, 2]$. The arrows on $x = 2$ show its outward energy flux during $1 < t < 3$.

Both the energy density and the flux are the same nonnegative function:

$$\rho_{\mathrm{en}}(t,x) = f^1(t,x) = F'(x-t)^2, \qquad F'(s)^2 = \begin{cases} 36s^2(1-s^2)^4, & |s| < 1, \\ 0, & |s| \geq 1. \end{cases} \tag{3.9}$$

The flux is positive for $0 < |x - t| < 1$ and zero otherwise. The total energy is an elementary integral:

$$E_{\mathrm{en},\mathbb{R}} = 36 \int_{-1}^{1} s^2(1-s^2)^4 \, \mathrm{d}s = 72 \left(\frac{1}{3} - \frac{4}{5} + \frac{6}{7} - \frac{4}{9} + \frac{1}{11} \right) = \frac{1024}{385}.$$

To compute the energy remaining in I, use the polynomial primitive

$$P(s) = 12s^3 - \frac{144}{5}s^5 + \frac{216}{7}s^7 - 16s^9 + \frac{36}{11}s^{11}, \qquad P'(s) = 36s^2(1-s^2)^4.$$

Its endpoint values are $P(1) = 512/385$ and $P(-1) = -512/385$. For $0 \leq t \leq 1$ the whole pulse lies in I. For $1 \leq t \leq 3$, substitution $s = x - t$ leaves the part $-1 \leq s \leq 2 - t$ in I. For $t \geq 3$, the interval contains no energy. Consequently

$$E_{\mathrm{en},I}(t) = \begin{cases} 1024/385, & 0 \leq t \leq 1, \\ P(2-t) + 512/385, & 1 \leq t \leq 3, \\ 0, & t \geq 3. \end{cases} \tag{3.10}$$

The endpoint flux confirms this result independently. At the left endpoint, $f^1(t,-2) = F'(-2-t)^2 = 0$ for all $t \geq 0$. At the right endpoint, $f^1(t,2) = F'(2-t)^2$. Thus

$$\mathcal{F}_I(t) = f^1(t,2) - f^1(t,-2) = F'(2-t)^2.$$

For $1 < t < 3$, differentiating (3.10) gives $E'_{\mathrm{en},I}(t) = -P'(2-t) = -\mathcal{F}_I(t)$. Outside this time interval both sides vanish. The total outward flux is

$$\int_0^3 \mathcal{F}_I(t) \, \mathrm{d}t = \int_1^3 F'(2-t)^2 \, \mathrm{d}t = \int_{-1}^{1} F'(s)^2 \, \mathrm{d}s = \frac{1024}{385} = E_{\mathrm{en},I}(0) - E_{\mathrm{en},I}(3).$$

The field conserves its total energy while the interval loses all its energy. This example guards the boundary hypothesis: local conservation does not imply constant energy in a fixed region. Lecture 5 uses boundary flux to study propagation. Lecture 13 uses outgoing flux to measure radiation, and Lecture 16 must distinguish energy in a region from total energy.

3.4 Variation of a background metric

The scalar tensor also describes how the action changes when the metric changes. This description will be used for the gravitational field in Lecture 28. Here the metric is prescribed, and the scalar field stays fixed during the variation.

Let $g_{\mu\nu}(x)$ be a C^1 symmetric nondegenerate matrix of signature $(+,-,\dots,-)$, with inverse $g^{\mu\nu}$. Write $g = \det(g_{\mu\nu})$ and $\mathrm{vol}_g = \sqrt{|g|}\,\mathrm{vol}$. On a compact coordinate region D the scalar action is

$$S_D[\phi, g] = \int_D L_g \,\mathrm{vol}_g, \qquad L_g = \frac{1}{2} g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - V(\phi).$$

We vary the inverse metric by a symmetric C^1 array $k^{\mu\nu}$ with compact support in the interior of D. For sufficiently small ε, $g^{\mu\nu} + \varepsilon k^{\mu\nu}$ remains nondegenerate with the same signature.

Definition 3.4 (Hilbert stress tensor)**.** The *Hilbert stress tensor* is the continuous symmetric array $T^{\mathrm{H}}_{\mu\nu}$ defined by the response to inverse-metric variation:

$$\delta_g S_D[\phi, g]k = \frac{1}{2}\int_D T^{\mathrm{H}}_{\mu\nu} k^{\mu\nu}\,\mathrm{vol}_g. \tag{3.11}$$

The variations range over symmetric arrays. The factor $\frac{1}{2}$ is part of the convention, including for off-diagonal entries. If two continuous symmetric arrays give the same response for every such variation, apply the fundamental lemma of Lecture 1 to each diagonal entry and to each symmetric pair of entries. Their difference is zero, so this definition determines at most one tensor.

Lemma 3.5 (Scalar metric variation)**.** *The scalar action has Hilbert tensor*

$$T^{\mathrm{H}}_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi - g_{\mu\nu}L_g. \tag{3.12}$$

At $g_{\mu\nu} = \eta_{\mu\nu}$ it agrees with the lowered canonical tensor. The equality holds for every field, without the field equation.

Proof. We differentiate the inverse and the determinant before differentiating the action. Differentiation of $g_{\mu\alpha}g^{\alpha\nu} = \delta^\nu_\mu$ gives

$$\delta g_{\mu\nu} = -g_{\mu\alpha}g_{\nu\beta}k^{\alpha\beta}.$$

For an invertible matrix G and an increment H, the determinant identity $\det(G + \varepsilon H) = \det G \det(\mathrm{id} + \varepsilon G^{-1}H)$, where id is the identity matrix, gives $\delta \det G = \det G\ \mathrm{tr}(G^{-1}H)$. Every non-identity permutation moves at least two indices, so its term in $\det(\mathrm{id} + \varepsilon G^{-1}H)$ has at least two factors of ε. The linear part of the identity term selects one diagonal entry of $\varepsilon G^{-1}H$ at a time. Their sum is $\varepsilon\,\mathrm{tr}(G^{-1}H)$. Applying this identity and then the inverse variation yields

$$\delta\sqrt{|g|} = \frac{1}{2}\sqrt{|g|}\,g^{\mu\nu}\delta g_{\mu\nu} = -\frac{1}{2}\sqrt{|g|}\,g_{\mu\nu}k^{\mu\nu}.$$

The absolute value causes no additional sign because g keeps its sign for small variations. Also, $\delta L_g = \frac{1}{2}\partial_\mu\phi\,\partial_\nu\phi\,k^{\mu\nu}$. The product rule therefore gives

$$\delta_g S_D[\phi, g]k = \frac{1}{2}\int_D \bigl(\partial_\mu\phi\,\partial_\nu\phi - g_{\mu\nu}L_g\bigr)k^{\mu\nu}\,\mathrm{vol}_g.$$

This is (3.11) with (3.12). Substitution of $g_{\mu\nu} = \eta_{\mu\nu}$ gives (3.3). ∎

For a variation $h_{\mu\nu} = \delta g_{\mu\nu}$ of the metric itself, $k^{\mu\nu} = -g^{\mu\alpha}g^{\nu\beta}h_{\alpha\beta}$. The equivalent convention is

$$\delta_g S_D = -\frac{1}{2}\int_D (T^{\mathrm{H}})^{\mu\nu}h_{\mu\nu}\,\mathrm{vol}_g.$$

Thus the functional derivative with respect to $g_{\mu\nu}$ carries a minus sign, whereas the derivative with respect to $g^{\mu\nu}$ carries the plus sign in (3.11).

Summary. For the autonomous scalar density, translations give $T^\mu{}_\nu = \partial^\mu\phi\,\partial_\nu\phi - \delta^\mu_\nu L$. On solutions its divergence vanishes. The energy density is $T^0{}_0$, and its spatial flux is $T^{i0} = -\phi_t\partial_i\phi$. The energy in a fixed rectangle changes by minus the net outward flux. Periodicity cancels that flux on the torus. Inverse-metric variation gives the same lowered symmetric tensor for the scalar field. Lecture 4 constructs Klein–Gordon solutions with finite energy.

Self-check. For a left-moving massless field $\phi(t, x) = F(x + t)$, compute the energy density and energy flux. Which sign has the flux?

Answer. Here $\phi_t = \phi_x = F'(x + t)$. Thus $\rho_{\mathrm{en}} = F'(x + t)^2$ and $f^1 = -F'(x + t)^2 \le 0$. Energy flows toward decreasing x.

Self-check. For the pulse of Section 3.3, compute the energy in I at $t = 2$ and the outward flux at that instant. Does a zero instantaneous flux imply that the interval's energy is constant over a time interval?

Answer. Since $P(0) = 0$, the remaining energy is $512/385$. The flux is $F'(0)^2 = 0$. It is positive at nearby times other than $t = 2$, so the energy is not constant on any neighbourhood of that time. Zero flux at one instant only gives zero derivative at that instant.

Self-check. Replace the autonomous scalar potential by $W(x, \phi)$, with W of class C^2. Compute $\partial_\mu T^\mu{}_\nu$ and identify when spatial translation conservation can fail even on a solution.

Answer. The chain rule adds the explicit derivative $-\partial_\nu^{\mathrm{explicit}}W$ to $\partial_\nu L$. Consequently $\partial_\mu T^\mu{}_\nu = -E[\phi]\partial_\nu\phi + \partial_\nu^{\mathrm{explicit}}W$. Spatially varying W can therefore give a nonzero divergence on a solution. Time-independent W still has zero explicit time derivative, so the energy balance survives.

Self-check. At a point where $k^{\mu\nu} = \lambda g^{\mu\nu}$, compute the pointwise volume variation $\delta\mathrm{vol}_g$ in $1 + d$ dimensions.

Answer. The contraction is $g_{\mu\nu}k^{\mu\nu} = \lambda(d + 1)$. Hence $\delta\mathrm{vol}_g = -\frac{1}{2}\lambda(d + 1)\mathrm{vol}_g$. In four spacetime dimensions the answer is $-2\lambda\mathrm{vol}_g$.